

ATKA, AK, USA

WMO: 700631

Lat: 52.220N

Lon: 174.206W

Elev: 56

StdP: 14.67

Time Zone: -10.00 (NAH)

Period: 06-19

WBAN: 25715

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 22.9	(c) 25.1	(d) 11.8	(e) 10.0	(f) 25.8	(g) 14.4	(h) 11.4	(i) 26.9	(j) 40.9	(k) 38.1	(l) 37.1	(m) 35.4	(n) 5.8	(o) 330	(p) 0.881

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 7.1	(c) 58.6	(d) 55.1	(e) 56.4	(f) 52.9	(g) 54.8	(h) 51.8	(i) 55.5	(j) 58.1	(k) 53.5	(l) 55.0	(m) 52.5	(n) 54.3	(o) 10.9	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
54.3	62.8	57.8	52.4	58.5	55.5	51.5	56.6	54.5	23.7	58.3	22.2	55.9	21.5	53.9	66.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 35.9	(b) 32.4	(c) 28.4	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 35.9	(b) 32.4	(c) 28.4	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.1	33.1	34.8	34.9	37.2	40.6	45.2	49.1	51.9	48.9	43.5	39.2	34.8	
	DBStd	7.16	4.29	4.22	4.45	3.24	2.98	2.45	2.83	2.65	3.00	3.73	3.72	4.01	
	HDD50	3349	524	426	467	384	293	145	50	9	54	202	324	471	
	HDD65	8712	989	846	932	834	758	594	493	407	483	666	774	936	
	CDD50	113	0	0	0	0	0	1	22	67	21	2	0	0	
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	
Wind	WSAvg	13.8	14.1	16.5	15.4	14.7	13.1	12.2	11.5	11.1	13.5	14.3	15.3	14.7	
Precipitation	PrecAvg	54.0	5.4	4.7	4.8	3.1	3.0	2.8	4.1	4.7	4.8	5.2	5.6	5.9	
	PrecMax	69.6	8.9	7.5	9.3	5.4	5.7	4.3	6.3	7.7	6.8	8.3	9.0	8.6	
	PrecMin	44.2	2.8	2.3	2.4	2.0	1.7	1.7	1.3	2.5	2.8	2.8	3.4	3.1	
	PrecStd	5.0	1.4	1.1	1.5	0.7	0.8	0.6	1.2	1.3	1.1	1.4	1.3	1.1	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.3	46.5	46.4	48.2	52.1	55.3	61.0	63.5	58.1	54.7	48.9	45.7	
		MCWB	39.9	41.4	40.9	40.8	45.2	50.4	56.6	58.7	53.8	51.3	46.1	42.1	
	2%	DB	41.3	45.0	44.9	45.3	47.7	52.3	56.8	60.6	55.3	52.3	47.5	44.6	
		MCWB	38.5	40.7	40.1	40.5	43.1	47.7	53.1	57.0	51.8	48.8	45.0	41.0	
	5%	DB	39.4	42.8	43.0	43.3	45.8	50.1	54.7	57.1	54.5	50.3	45.9	42.9	
		MCWB	36.9	39.6	38.8	39.1	41.9	46.5	51.5	53.9	51.2	47.0	43.3	39.6	
	10%	DB	38.8	40.2	40.9	42.5	45.1	48.2	53.7	55.2	53.8	48.3	45.1	40.7	
		MCWB	36.2	37.5	37.5	38.6	41.3	45.4	50.9	52.3	50.7	45.5	42.2	37.9	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.1	43.6	43.4	43.2	46.5	50.3	57.3	59.3	55.2	51.5	47.2	43.0
			MCDB	43.3	45.7	46.0	46.1	49.0	53.2	61.2	62.7	57.9	53.5	48.3	45.0
2%		WB	39.3	41.3	40.9	41.3	44.2	48.0	53.9	57.0	53.1	49.3	46.0	41.2	
		MCDB	41.5	43.8	43.8	44.4	46.0	50.5	56.1	60.0	54.9	51.6	46.9	44.0	
5%		WB	38.1	39.5	39.6	40.2	42.5	46.5	51.6	54.6	51.5	47.1	44.3	39.9	
		MCDB	40.2	42.6	42.7	43.4	45.2	48.3	54.2	56.5	53.8	49.3	45.8	42.7	
10%		WB	37.0	38.0	38.0	38.9	41.4	45.9	50.4	53.2	50.4	46.0	42.4	38.3	
		MCDB	39.0	40.4	40.9	41.9	44.7	47.4	52.8	54.7	52.8	47.9	44.9	40.7	
Mean Daily Temperature Range		5% DB	MDBR	6.3	6.1	6.7	6.7	6.4	6.2	6.6	7.1	6.5	6.7	6.6	6.9
			MCDBR	6.3	8.3	8.7	8.6	8.2	9.7	9.9	10.6	8.4	7.7	8.0	8.4
	MCWBR		6.0	8.6	7.5	7.2	5.9	6.2	7.0	8.1	6.7	6.7	7.8	7.5	
	5% WB	MCDBR	5.9	7.9	8.5	8.3	7.1	7.7	9.2	9.9	7.6	6.9	8.0	7.4	
		MCWBR	6.1	8.4	7.8	7.5	5.7	5.7	6.8	8.1	6.5	6.6	8.8	7.3	
Clear-Sky Solar Irradiance	taub	0.311	0.340	0.364	0.417	0.413	0.415	0.417	0.401	0.386	0.356	0.346	0.301		
	taud	2.327	2.282	2.262	2.108	2.188	2.262	2.316	2.368	2.314	2.297	2.178	2.308		
	Ebn at Noon	215	242	260	258	266	266	264	262	250	232	190	197		
	Edh at Noon	20	28	34	45	43	41	38	34	32	27	23	17		
All-Sky Solar Radiation	RadAvg	213	409	769	1149	1255	1254	1190	1041	865	564	271	173		
	RadStd	21	44	61	108	103	119	87	87	60	53	26	18		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page